

Bryle Jaron Ong

858-705-0531 | ongbrylejaron@gmail.com | linkedin.com/in/brylejaronong | github.com/brylejaronong

EDUCATION

San Diego State University — *B.S. Computer Science, B.A. Philosophy*

Graduated Spring 2026

GPA: 3.86 — *Relevant Classes: Data Structures & Algorithms, Operating Systems, Networks & Distributed Systems, UNIX System Administration*

SKILLS

Languages: Python, C/C++, Java, HTML/CSS, JavaScript, MATLAB, MIPS

Developer Tools: Git, Github, Vim, VSCode, Bash

WORK EXPERIENCE

Autonomous Systems Engineer — *Real-Time Embedded & Controls (RTEC) Lab*

Aug. 2025 – May 2026

- Deployed **ROS2** autonomous stack (AMCL, global planning, Pure Pursuit) fusing LiDAR, IMU, and odometry
- Built graph-based local planner enabling dynamic **overtaking**, one of two planning approaches on the team
- Coordinated integration and sprint workflow for a 5-person team via **Jira**

Student Researcher — *AI4Business Lab*

Feb. 2024 – May 2024

- Conducted a **literature review** on LLMs for code-related tasks as part of a multi-person research team
- Summarized model architectures including encoder, decoder, and encoder-decoder designs
- Explored open challenges including **long-context understanding** and computational **resource constraints**

Service & Food Champion — *Taco Bell*

Jul. 2022 – Dec. 2023

- Packaged food accurately and quickly, following guidelines for **food safety** and kitchen equipment
 - Assisted customers in the drive-thru and dine-in via **order-taking** with smooth operation
 - Communicated openly with coworkers and managers, following principles of **teamwork** and **collaboration**
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LEADERSHIP EXPERIENCE

Robotics Mentor — *FIRST Robotics*

Jan. 2022 – May 2022

- Co-mentored an **FLL** team of 5 students, guiding middle schoolers to autonomously deliver a package
 - Volunteered ~2 hours/week over 4 months, supporting hands-on robotics and programming skill-building
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PROJECTS

Two-Pass Assembler — *SIC/XE Architecture*

C++ | Makefile | Git

- Supported instruction **formats 1-4**, **immediate/indirect/indexed** addressing, and **PC-relative/base-relative**
- Designed modular architecture separating **lexing**, **SYMTAB**, and **OPTAB** using a hash-map for O(1) lookups
- Collaborated with teammate via Git, dividing work by assembler pass while sharing intermediate file

Restaurant Seating Simulator — *Producer-Consumer System*

C++ | POSIX Thread

- Implemented a multithreaded system using a **bounded buffer** shared between producer/consumer
- Synchronized with pthread **mutexes** and condition variables to prevent race conditions
- Used a **semaphore** to coordinate graceful shutdown once all producers finish, ensuring no lost requests

P2P Chat — *Peer-to-Peer Encrypted Group Chat*

Python | WebSockets | RSA

- Discovered peers via **UDP** broadcast on **LAN**, establishing direct **TCP** connections
- Implemented per-peer **RSA-2048** key exchange and **OAEP** encryption with handshake logic
- Combined multithreaded networking backend with an **asyncio**-based web UI bridge

Calculated Collapse — *Multiplayer FPS*

Unity | C# | Blender

- Co-developed a 1v1 multiplayer FPS in **Unity** with a 5-person team, featuring pre-round phase and combat
- Programmed dash and slide mechanics, including **input handling**, **movement logic**, and **physics values**
- Implemented networked **animations** and **sound effects**, and rigged character models using Blender